MILITARY SPECIFICATION

RESISTOR NETWORKS, FIXED, FILM, ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for nonhermetically sealed resistor networks that possess a high degree of stability with respect to time under severe environmental conditions, with an established reliability. Resistors covered by this specification will have life failure rates ranging from 1 percent to 0.001 percent per 1,000 hours (see 1.2.1.5). These failure rates are established at 60 percent confidence on the basis of life tests. The failure rate level identified by appropriate symbol, is referred to operation at full rated voltage and rated temperature, with a permissible change in resistance of 2 percent as the criteria for failure. A part per million (ppm) quality system is used for documenting and reporting the average outgoing quality of resistors supplied to this specification. Statistical Process Control (SPC) techniques are required in the manufacturing process to minimize variation in production of resistors supplied to the requirements of this specification.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). Networks specified herein (see 3.1), shall be identified by a PIN which shall consist of the basic number of the associated detail specification and a coded number. Each associated detail specification covers a different network style. The number shall be coded to provide information concerning the network characteristic, resistance value, resistance tolerance and schematic. The PIN shall be in the following form with a coded number derived as indicated:

M874	H	01	F	1002	A	S
Military specification number (indicating MIL-R-874)	Characteristic (see 1.2.1.1)	Military specification sheet number (indicating MIL-R-874/1)	Resistance tolerance (see 1.2.1.2)	Resistance (see 1.2.1.3	Schematic (see 1.2.1.4	Failure rate (see 1.2.1.5)
	Coded dash number					

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905

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1.2.1.1 Characteristic. The characteristic is identified by the single letters H, K, or M in accordance with table I.

TABLE I. Characteristics.

		H	K	M	
Resistance-temperature characteristic (see 3.13)		±50	±100	±300	ppm/°C
Maximum ambient temperature at rated wattage (3.6)		70	70	70	°C
Maximum ambient temperature at zero power derating (see figure 2)		125	125	125	
Thermal shock (see 3.8) and Power conditioning (see 3.9)	ΔR ΔRatio	±.5	±.7	±.7	Maximum percent change in resistance (0.01 ohm additional allowed for measurement error) and, when applicable, maximum percent change in resistance ratio
Low temperature operation (see 3.14)	ΔR ΔRatio	±.1	±.25	±.5	
Short-time overload (see 3.15)	ΔR ΔRatio	±.1	±.25	±.5	
Terminal strength (see 3.16)	ΔR ΔRatio	±.25	±.25	±.25	
Resistance to soldering heat (see 3.19)	ΔR ΔRatio	±.1	±.25	±.25	
Moisture resistance (see 3.20)	ΔR ΔRatio	±.4	±.5	±.5	
Shock (specified pulse) (see 3.21)	ΔR ΔRatio	±.25	±.25	±.25	
Vibration, high frequency (see 3.22)	ΔR ΔRatio	±.25	±.25	±.25	
Life (see 3.23)	ΔR ΔRatio	±.5	±.5	±2.0	
Power ratings 25°C (see 3.23.1)	ΔR ΔRatio	±.5	±.5	±2.0	
High temperature exposure (see 3.24)	ΔR ΔRatio	±.2	±.5	±1.0	
Low temperature storage (see 3.25)	ΔR ΔRatio	±0.1	±.25	±.5	
Insulation resistance (see 3.18)		10,000	10,000	10,000	
Resistance tolerance and, when applicable, resistance ratio accuracy (see table IV)		0.1(B) 0.5(D) 1.0(F)	0.5(D) 1.0(F) 2.0(G) 5.0(J)	1.0(F) 2.0(G) 5.0(J)	

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TABLE II. Resistance tolerance.

Symbol	Resistance tolerance (percent)
B	0.10
D	0.50
F	1.0
G	2.0
J	5.0

1.2.1.2 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table II.

1.2.1.3 Resistance.

1.2.1.3.1 Four digit resistance designation (all resistor values equal). The four digit resistance designation is applicable to all resistance tolerances. The nominal resistance is identified by four digits; the first three digits represent significant figures and the last digit specifies the number of zeros to follow. When the value of resistance is less than 100 ohms, or when fractional values of an ohm are required, the letter R shall be substituted for one of the significant digits to represent the decimal point. When the letter R is used, succeeding digits of the group represent significant figures. The resistance value designations are shown in table III. Standard values for every decade shall follow the sequence demonstrated for "10 to 100" decade in table IV. Resistance values not listed in table IV for the appropriate resistance tolerance shall be considered as not conforming to the specification. The resistance values for tolerance B may be any value within the limits specified in 3.1, but it is preferred that values be chosen from the "D" column of table IV.

Although resistance tolerances G and J normally require less than three significant figures to adequately describe the true resistance value, for the purpose of this specification, the nominal value shall be three significant figures followed by the fourth digit to signify the number of zeros to follow.

TABLE III. Designation of resistance values for resistance tolerances of .005, .01, .05, 0.10, 0.50, 1.0, 2.0, and 5.0 percent.

Designation	Resistance (ohms)
1R00 to 9R88 inclusive	1.00 to 9.88 inclusive
10R0 to 98R8 inclusive	10.0 to 98.8 inclusive
1000 to 9880 inclusive	100 to 988 inclusive
1001 to 9881 inclusive	1,000 to 9,880 inclusive
1002 to 9882 inclusive	10,000 to 98,800 inclusive
1003 to 9883 inclusive	100,000 to 988,000 inclusive
1004 to 9884 inclusive	1,000,000 to 9,880,000 inclusive

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TABLE IV. Standard resistance values for the 10 to 100 decade for 0.1, 0.5, 1.0, 2.0, and 5.0 percent resistance tolerance.

0.1 (B) 0.5 (D)	1.0 (F)	2.0 (G) 5.0 (J)	0.1 (B) 0.5 (D)	1.0 (F)	2.0 (G) 5.0 (J)	0.1 (B) 0.5 (D)	1.0 (F)	2.0 (G) 5.0 (J)
10.00	10.00	10.00	22.90	---	---	49.30	---	---
10.10	---	---	23.20	23.20	---	49.90	49.90	---
10.20	10.20	---	23.40	---	---	50.50	---	---
10.40	---	---	23.70	23.70	---	---	---	51.00
10.50	10.50	---	24.00	---	24.00	51.10	51.10	---
10.60	---	---	24.30	24.30	---	51.70	---	---
10.70	10.70	---	24.60	---	---	52.30	52.30	---
10.90	---	---	24.90	24.90	---	53.00	---	---
11.00	11.00	11.00	25.20	---	---	53.60	53.60	---
11.10	---	---	25.50	25.50	---	54.20	54.20	---
11.30	11.30	---	25.80	---	---	54.90	54.90	---
11.40	---	---	26.10	26.10	---	55.60	---	---
11.50	11.50	---	26.40	---	---	---	---	56.00
11.70	---	---	26.70	26.70	---	56.20	56.20	---
11.80	11.80	---	---	---	27.00	56.90	---	---
12.00	---	12.00	27.10	---	---	57.60	57.60	---
12.10	12.10	---	27.40	27.40	---	58.30	---	---
12.30	---	---	27.70	---	---	59.00	59.00	---
12.40	12.40	---	28.00	28.00	---	59.70	---	---
12.60	---	---	28.40	---	---	60.40	60.40	---
12.70	12.70	---	28.70	28.70	---	61.20	---	---
12.90	---	---	29.10	---	---	61.90	61.90	---
13.00	13.00	13.00	29.40	29.40	---	---	---	62.00
13.20	---	---	29.80	---	---	62.60	---	---
13.30	13.30	---	---	---	30.00	63.40	63.40	---
13.50	---	---	30.10	30.10	---	64.20	---	---
13.70	13.70	---	30.50	---	---	64.90	64.90	---
13.80	---	---	30.90	30.90	---	65.70	---	---
14.00	14.00	---	31.20	---	---	66.50	66.50	---
14.20	---	---	31.60	31.60	---	67.30	---	---
14.30	14.30	---	32.00	---	---	---	---	68.00
14.50	---	---	32.40	32.40	---	68.10	68.10	---
14.70	14.70	---	32.80	---	---	69.00	---	---
14.90	---	---	---	---	33.00	69.80	69.80	---
15.00	15.00	15.00	33.20	33.20	---	70.60	---	---
15.20	---	---	33.60	---	---	71.50	71.50	---
15.40	15.40	---	34.00	34.00	---	72.30	---	---
15.60	---	---	34.40	---	---	73.20	73.20	---
15.80	15.80	---	34.80	34.80	---	74.10	---	---
16.00	---	16.00	35.20	---	---	75.00	75.00	75.00
16.20	16.20	---	35.70	35.70	---	75.90	---	---
16.40	---	---	---	---	36.00	76.80	76.80	---
16.50	16.50	---	36.10	---	---	77.70	---	---
16.70	---	---	36.50	36.50	---	78.70	78.70	---
16.90	16.90	---	37.00	---	---	79.60	---	---
17.20	---	---	37.40	37.40	---	80.60	80.60	---
17.40	17.40	---	37.90	---	---	81.60	---	---
17.60	---	---	38.30	38.30	---	---	---	82.00
17.80	17.80	---	38.80	---	---	82.50	82.50	---
18.00	18.00	18.00	---	---	39.00	83.50	---	---
18.20	18.20	---	39.20	39.20	---	84.50	84.50	---
18.40	---	---	39.70	---	---	85.60	---	---
18.70	18.70	---	40.20	40.20	---	86.60	86.60	---
18.90	---	---	40.70	---	---	87.60	---	---

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TABLE IV. Standard resistance values for the 10 to 100 decade for 0.1, 0.5, 1.0, 2.0, and 5.0 percent resistance tolerance - Continued.

0.1 (B) 0.5 (D)	1.0 (F)	2.0 (G) 5.0 (J)	0.1 (B) 0.5 (D)	1.0 (F)	2.0 (G) 5.0 (J)	0.1 (B) 0.5 (D)	1.0 (F)	2.0 (G) 5.0 (J)
19.10	19.10	---	41.20	41.20	---	88.70	88.70	---
19.30	---	---	41.70	---	---	89.80	---	---
19.60	19.60	---	42.20	42.20	---	90.90	90.90	---
19.80	---	---	42.70	---	---	---	---	91.00
20.00	20.00	20.00	---	---	43.00	92.00	---	---
20.30	---	---	43.20	43.20	---	93.10	93.10	---
20.50	20.50	---	43.70	---	---	94.20	---	---
20.80	---	---	44.20	44.20	---	95.30	95.30	---
21.00	21.00	---	44.80	---	---	96.50	---	---
21.30	---	---	45.30	45.30	---	97.60	97.60	---
21.50	21.50	---	45.90	---	---	98.80	---	---
21.80	---	---	46.40	46.40	---			
---	---	22.00	47.00	---	47.00			
22.10	22.10	---	47.50	47.50	---			
22.30	---	---	48.10	---	---			
22.60	22.60	---	48.70	48.70	---			

1.2.1.3.2 Resistance designation for network schematics "H" and "J". For network schematics "H" and "J", the four digit resistance designation becomes a code indicator with the first digit being an "A" and the following three digits a code number indicating a resistance combination described in the applicable associated military specification.

1.2.1.4 Schematic. The network schematic shall be identified by a single letter in accordance with figure 1, and as specified (see 3.1). Dotted lines in the schematics refer to configurations that might have additional resistors.

1.2.1.5 Life failure rate designation. The life failure-rate designation as shown in table V is signified by a single letter (M, P, R, S), which identifies the life failure-rate level for which the resistor is qualified (see 4.4.4 and 6.3).

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TABLE V. Life failure rate level (established at 60 percent confidence).

Failure rate level designation	Failure rate percent/1,000 hours
M	1.0
P	0.1
R	0.01
S	0.001

1.2.2 Style. The style is identified by the three-letter symbol "RZR" followed by a two-digit number (see 3.1). The letters identify fixed resistor networks and the two digits represent envelope size and configuration.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifictions and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder; Tin Alloy; Lead-Tin alloy; and Lead Alloy.

MILITARY

MIL-R-39032 - Resistors, Packaging of.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-690 - Failure Rate Sampling Plans and Procedures.
MIL-STD-790 - Reliability Assurance Program For Electronic Specifications.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-45662 - Calibration Systems Requirements.

(See supplement 1 for list of associated detail specifications.)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION

EIA-554 - Assessment of Outgoing Noncomforming Levels in Part per Million (PPM).
EIA-557 - Statistical Process Control Systems.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document shall takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specifications. The individual part requirements shall be as specified herein and in accordance with the applicable associated detail specifications. In the event of any conflict between requirements of this specification and the associated detail specifications, the latter shall govern (see 6.2).

3.2 Qualification. Networks furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Reliability and quality.

3.3.1 Reliability. Reliability of resistors furnished under this specification shall be established and maintained in accordance with procedures and requirements specified in MIL-STD-790 and MIL-STD-690 with details and exceptions specified in 4.1.3, 4.4.4, and 4.5.

3.3.2 Quality.

3.3.2.1 Statistical process control (SPC). The contractor shall implement and use statistical process control techniques in the manufacturing process for parts covered by this specification. The SPC program shall be developed and maintained in accordance with EIA-557. The SPC program shall be documented and maintained as part of the overall reliability assurance program as specified in MIL-STD-790. The implementation of statistical process control shall be 12 months from the date of the specification. Process for application of SPC techniques should include but are not limited to:

a. Printing of resistance elements.

b. Laser trimming.

c. Termination finish.

d. Encapsulation.

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3.3.2.2 Quality levels. The quality of lots that have been subjected to and have passed the subgroup 1, 100 percent screening inspections of group A inspection shall be established and maintained in accordance with 4.4.4 and EIA-554. Individual ppm defect levels (i.e., ppm-2 and ppm-3) and an overall ppm defect level (i.e., ppm-5) shall be established based on the tests prescribed in the subgroup 2 tests of the group A inspections. The defect for ppm-2 shall be less than 100 ppm. Data shall not be excluded from the appropriate ppm calculation unless specifically authorized by the qualifing activity. Guidance for exclusion of data is specified in EIA-554.

3.3.2.2.1 Noncompliance. The contractor shall notify the qualifying activity when the 100 ppm level is reached or exceeded for ppm-2. The contractor shall provide sufficient information to the qualifying activity documenting the causes of the problem and what corrective action is being taken. Failure to correct this problem shall be the basis for removal of the affected product from the QPL.

3.4 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the networks to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.5 Design and construction. Networks shall be of the design, construction, and physical dimensions specified (see 3.1). Each network shall consist of film type resistance elements with terminations or film type resistance elements (see 3.1).

3.5.1 Enclosure. Networks shall be encapsulated sufficiently to withstand the environmental tests specified. There shall be no voids which expose the internal circuitry.

3.5.2 Termination leads. Terminal leads shall be free of foreign material and solderable as specified. Leads shall be judged to be free of foreign material if the visual criteria are met.

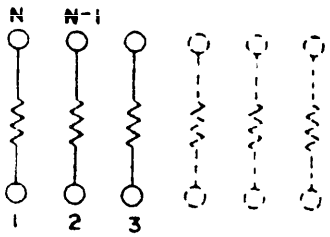
3.5.3 Metallization resistance. The resistance of the metallization of the longest path in the network shall not exceed the limit in table VI.

TABLE VI. Metallization resistance.

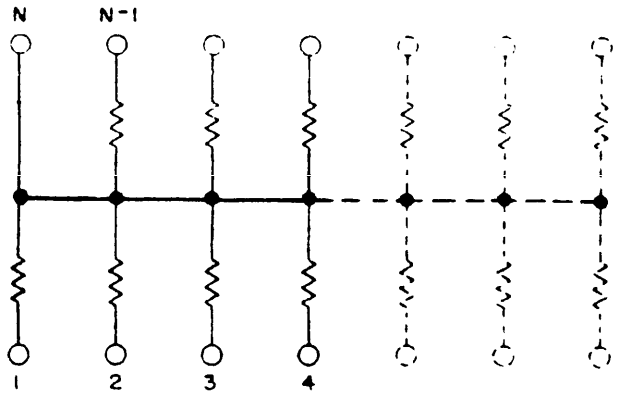
Nominal element resistance value (ohms) R	Maximum metallization resistance (ohms)
R $\leq$ 1,000	1
1,000 $\leq$ R $\leq$ 10,000	5
R $\geq$ 10,000	10

3.5.4 Soldering. If soldering is used for internal connections, it shall be made with solder, in accordance with QQ-S-571, having a liquid point not less than +280°C.

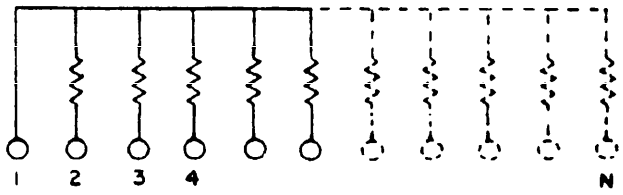
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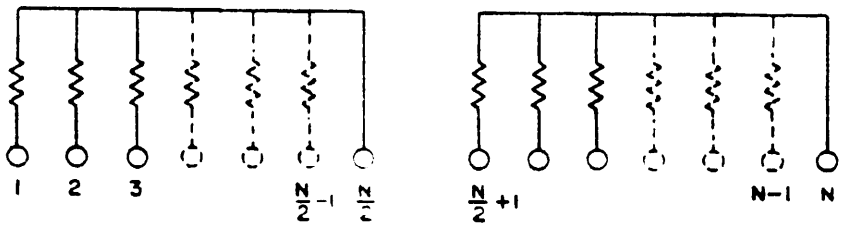
SCHEMATIC "A"



SCHEMATIC "B"



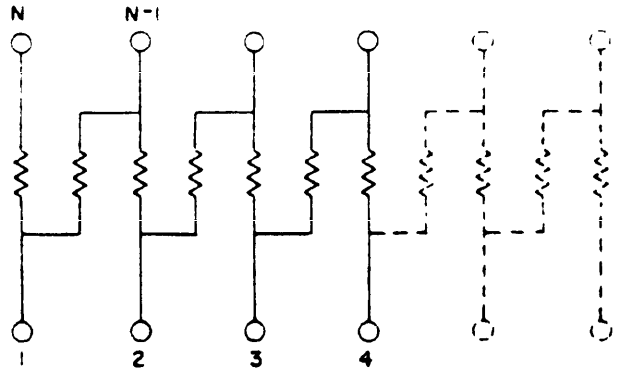
SCHEMATIC "C"



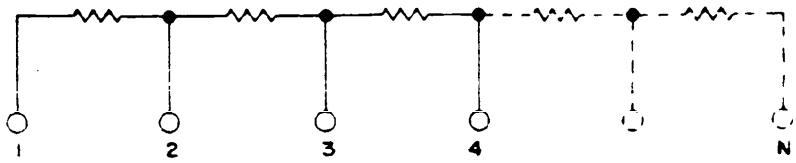
SCHEMATIC "D"

FIGURE 1. Schematics.

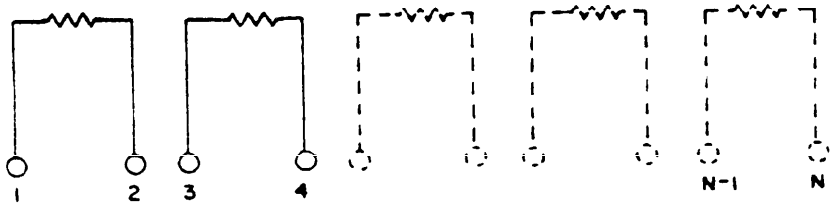
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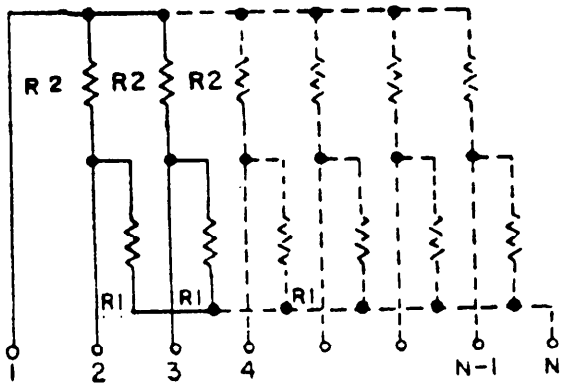
SCHEMATIC "E"



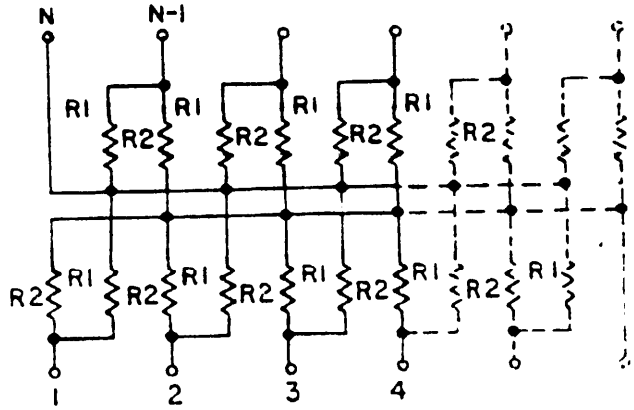
SCHEMATIC "F"



SCHEMATIC "G"



SCHEMATIC "H"



SCHEMATIC "J"

FIGURE 1. Schematics - Continued.

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3.6 Power rating. The networks and individual resistors shall have a power rating based on continuous full-load operation at an ambient temperature of 70°C. For temperatures other than 70°C, the power rating shall be in accordance with figure 2.

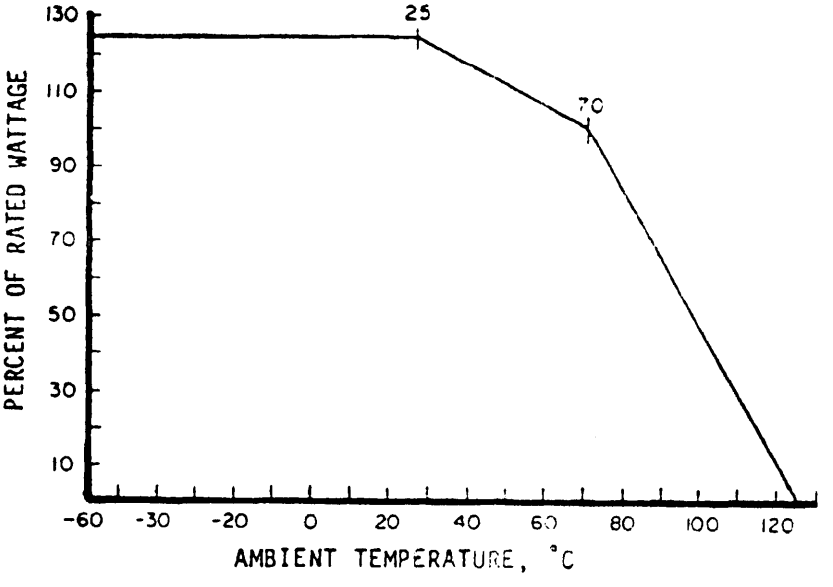


FIGURE 2. Power derating curve.

3.7 Voltage rating. Each resistor element shall have a rated dc continuous working voltage or an approximate sine-wave root-mean-square (rms) continuous working voltage corresponding to the wattage (power) rating, as determined from the following formula:

$$E = \sqrt{PR}$$

E = Continuous rated dc or rms working voltage in volts.
P = Rated wattage in watts.
R = Nominal resistance in ohms.

In no case shall the rated voltage be greater than the applicable maximum voltage (see 3.1).

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3.8 Thermal shock. When networks are tested as specified in 4.8.3, there shall be no evidence of mechanical damage. For the combined thermal shock and power conditioning test (see 3.9) of qualification inspection group I, and group A inspection, the networks shall meet the following requirements:

Characteristics K and M:	ΔR shall not exceed ±(0.7 percent +0.01 ohm).
Characteristic H:	ΔR shall not exceed ±(0.5 percent +0.01 ohm).

For the thermal shock test of qualification inspection group IV, and group C inspection, networks shall meet the following requirements:

Characteristics K and M:	ΔR shall not exceed ±(0.50 percent +0.01 ohm).
Characteristic H:	ΔR shall not exceed ±(0.25 percent +0.01 ohm).

3.9 Power conditioning. When networks are tested as specified in 4.8.4, there shall be no mechanical damage. For the combined power conditioning and thermal shock test (see 3.8), networks shall meet the following requirements:

Characteristics K and M:	ΔR shall not exceed ±(0.7 percent +0.01 ohm).
Characteristic H:	ΔR shall not exceed ±(0.5 percent +0.01 ohm).

3.10 DC resistance.

3.10.1 Individual resistance. When networks are tested as specified in 4.8.5, the dc resistance shall be within the specified tolerance of the nominal resistance (see 1.2.1.3) and shall be stable within the specified tolerance during the measurement.

3.10.2 Resistance between isolated pins. The resistance between any two isolated resistors tested as specified in 4.8.5 shall be not less than 5 megohms. This shall only be measured during the resistance test on group I of qualification and group A inspection.

3.11 Solderability. When networks are tested as specified in 4.8.6, they shall meet the criteria for wire lead evaluation in the test standard.

3.12 Resistance to solvents. When networks are tested as specified in 4.8.7, there shall be no evidence of mechanical damage and the marking shall remain legible.

3.13 Resistance-temperature characteristic. When networks are tested as specified in 4.8.8, the resistance-temperature characteristic, at each of the temperatures specified in table XVII referred to room ambient temperature, shall not exceed the following requirements:

Characteristic M:	±300 ppm.
Characteristic K:	±100 ppm.
Characteristic H:	±50 ppm.

3.14 Low temperature operation. When networks are tested as specified in 4.8.9, there shall be no evidence of mechanical damage and shall meet the following requirements:

Characteristic M:	ΔR shall not exceed ±(0.50 percent, +0.01 ohm).
Characteristic K:	ΔR shall not exceed ±(0.25 percent, +0.01 ohm).
Characteristic H:	ΔR shall not exceed ±(0.10 percent, +0.01 ohm).

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3.15 Short-time overload. When networks are tested as specified in 4.8.10, there shall be no evidence of arcing, burning, or charring. The networks shall meet the following requirements:

Characteristic M: ΔR shall not exceed ±(0.5 percent, +0.01 ohm).
Characteristic K: ΔR shall not exceed ±(0.25 percent, +0.01 ohm).
Characteristic H: ΔR shall not exceed ±(0.1 percent, +0.01 ohm).

3.16 Terminal strength. When networks are tested as specified in 4.8.11, there shall be no evidence of breaking or loosening of terminals from the network form, or chipping of coating. The networks shall meet the following requirements:

Characteristics M, K, and H: ΔR shall not exceed ±(0.25 percent, +0.01 ohm).

3.17 Dielectric withstanding voltage. When networks are tested as specified in 4.8.12, there shall be no mechanical damage, arcing, or breakdown. The leakage current shall not exceed 1 milliampere (mA).

3.18 Insulation resistance. When networks are tested as specified in 4.8.13, the insulation resistance shall not be less than 10,000 megohms.

3.19 Resistance to soldering heat. When networks are tested as specified in 4.8.14, there shall be no external evidence of mechanical damage nor internal evidence of solder reflow or heat damage. The networks shall meet the following requirements:

Characteristics K and M: ΔR shall not exceed ±(0.25 percent, +0.01 ohm).
Characteristic H: ΔR shall not exceed ±(0.1 percent, +0.01 ohm).

3.20 Moisture resistance. When networks are tested as specified in 4.8.15, there shall be no evidence of mechanical damage. The networks shall meet the following requirements:

Characteristics K and M: ΔR shall not exceed ±(0.5 percent, +0.01 ohm).
Characteristic H: ΔR shall not exceed ±(0.4 percent, +0.01 ohm).

Additionally, the dielectric withstanding voltage shall be as specified in 3.17 and the insulation resistance shall be 100 megohms minimum.

3.21 Shock (specified pulse). When networks are tested as specified in 4.8.16, there shall be no evidence of mechanical damage and they shall meet the following requirements:

Characteristics H, K, and M: ΔR shall not exceed ±(0.25 percent, +0.01 ohm).

3.22 Vibration, high frequency. When networks are tested as specified in 4.8.17, there shall be no mechanical damage and they shall meet the following requirements:

Characteristics H, K, and M: ΔR shall not exceed ±(0.25 percent, +0.01 ohm).

3.23 Life. When networks are tested as specified in 4.8.18, there shall be no evidence of mechanical damage and they shall meet the following requirements:

Characteristic M: ΔR shall not exceed ±(2.0 percent, +0.01 ohm).
Characteristics H and K: ΔR shall not exceed ±(0.5 percent, +0.01 ohm).

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3.23.1 Power ratings at 25°C (see 3.1 and 6.5). For qualification testing only, networks shall be subjected to the specified voltage and shall dissipate the assigned 25°C wattage rating for a 2000-hour duration. The networks shall meet the following requirements:

Characteristic M:	ΔR shall not exceed ±(2.0 percent, +0.01 onm).
Characteristics H and K:	ΔR shall not exceed ±(0.5 percent, +0.01 ohm).

3.23.2 Failure rate determination. When resistors are tested as specified in 4.8.18, there shall be no evidence of mechanical damage to the resistance element, or enclosure. The change in resistance between the initial measurement and any of the succeeding measurements shall not exceed ±(2.0 percent, +0.01 ohm). This single failure criteria shall be applicable to all measurements during the life test for purposes of determining failure rate level qualification, and is applicable as a parallel requirement with 3.23.1 to the measurements made during the life test specified for qualification inspection.

3.24 High temperature exposure. When networks are tested as specified in 4.8.19, there shall be no evidence of mechanical damage and they shall meet the following requirements:

Characteristic M:	ΔR shall not exceed ±(1.0 percent, +0.01 ohm).
Characteristic K:	ΔR shall not exceed ±(0.5 percent, +0.01 ohm).
Characteristic H:	ΔR shall not exceed ±(0.2 percent, +0.01 ohm).

3.25 Low temperature storage. When networks are tested as specified in 4.8.20, there shall be no evidence of mechanical damage and they shall meet the following requirements:

Characteristic M:	ΔR shall not exceed ±(0.5 percent, +0.01 ohm).
Characteristic K:	ΔR shall not exceed ±(0.25 percent, +0.01 ohm).
Characteristic H:	ΔR shall not exceed ±(0.1 percent, +0.01 ohm).

3.26 Fungus. All external materials shall be nonnutrient to fungus growth or shall be suitably treated to retard fungus growth. The manufacturer shall verify by certification that all external materials are fungus resistant or shall test the resistors as specified in 4.8.21. There shall be no evidence of fungus growth on the external surfaces.

3.27 Marking. Each network shall be marked in accordance with method I of MIL-STD-1285 and as indicated below.

a. Complete military Part or Identifying Number (PIN).

b. Manufacturer's CAGE code.

c. Resistance value and resistance tolerance (manufacturer's option).

d. Manufacturing date code.

e. Manufacturer's lot code (manufacturer's option).

f. Pin no. 1 identifier.

g. JAN marking.

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temperature for each of the succeeding temperatures

3.27.1 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN".

3.27.2 Failure rate substitution. A manufacturer may supply to all higher failure rate levels than to which he is qualified. Parts qualified and marked to lower failure rate levels, with procuring activity approval, are substitutable for parts marked to higher failure rate levels, and shall not be remarked unless specified in the contract or purchase order (see table VII).

3.27.3 Resistance temperature characteristics and resistance tolerances substitution. Networks qualified and marked to lower resistance temperature characteristics or lower tolerance levels, or both, with procuring agency approval, are substitutable for networks marked to higher resistance temperature characteristics or higher tolerance levels, and shall not be remarked unless specified in the contract or purchase order (see 6.1) (see tables VIII and IX).

TABLE VII. Failure rate substitution.

Failure rate	Failure rate substitution
S	S, R, P, M
R	R, P, M
P	P, M
M	M

TABLE VIII. Substitution of temperature characteristics.

Characteristic	Characteristic substitution
H	H, K, M
K	K, M
M	M

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may solder dip/reflow as follows:

TABLE IX. Resistance tolerance substitution.

Resistance tolerance	Resistance tolerance substitution
B	B, D, F, G, J
D	D, F, G, J
F	F, G, J
G	G, J
J	J

3.28 Workmanship. Networks shall be processed in such a manner as to be uniform in quality and shall meet the requirements of 3.4 to 3.5.4 inclusive, and shall be free from other defects that will affect life, serviceability, or appearance.

3.29 Resistance value deviations. All maximum deviations as specified in this section are to be considered absolute limits with the exception of the contact resistance adjustments.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.3 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790.

4.1.4 Statistical process control (SPC). A SPC program shall be established and maintained in accordance with EIA-557. Evidence of such compliance shall be verified by the qualifying activity as a prerequisite for qualification and retention of qualification.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.6).

c. Quality conformance inspection (see 4.7).

4.3 Inspection conditions and precautions.

4.3.1 Conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.2 Precautions. Adequate precautions shall be taken during inspection to prevent condensation of moisture on networks, except during moisture-resistance test.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample. The number of sample units comprising a sample of networks to be submitted for qualification inspection shall be as specified in the appendix of this specification. The sample shall be taken at random from a production run and shall be produced with equipment and procedures normally used in production. The sample units shall have been subjected to and passed the requirements of group A inspection (see 4.7.1.2). Qualification shall not be granted if group A inspection requirements are not met. Each network style shall be qualified separately (see 3.1).

4.4.2 Inspection routine. Sample units shall be subjected to the qualification inspection specified in table X in the order shown. All sample units shall be subjected to the inspection of groups I and IA, except for the sample units required for groups II and VIII. All sample units shall be divided as specified in table X for groups II through VIII inclusive, and subjected to the inspection for their particular group. For purposes of verifying the 25°C power rating (see 3.1 and 6.5), a separate sample of 30 units shall be subjected to group I, group IA, and group VIA of table X. This test is an initial qualification requirement only. Sample sizes and extent of qualification for characteristics shall be specified in the appendix of this specification.

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TABLE X. Qualification inspection.

Inspection	Number of sample units	Requirement paragraph	Method paragraph	Number of failures allowed 1/
Group I 2/3/				
Thermal shock	All sample units 4/	3.8	4.8.3	5/
Power conditioning		3.9	4.8.4	
DC resistance		3.10	4.8.5	
Group IA 3/6/				
Visual and mechanical inspection		3.1, 3.4, 3.5, 3.28, 3.29	4.8.2	0
Group II 7/				
Solderability	12 4/	3.11	4.8.6	1
Resistance to solvents		3.12	4.8.7	
Group III				
Resistance-temperature characteristic	20 10 high or 10 critical 8/ 30 10 low	3.13	4.8.8	1 9/
Low temperature operation		3.14	4.8.9	
Short-time overload		3.15	4.8.10	
Terminal strength		3.16	4.8.11	
Group IV				
Thermal shock	20 10 high or 10 critical 8/ 30 10 low	3.8	4.8.3	1 9/
Dielectric withstanding voltage		3.17	4.8.12	
Insulation resistance		3.18	4.8.13	
Resistance to soldering heat 10/		3.19	4.8.14	
Moisture resistance		3.20	4.8.15	
Group V				
Shock (specified pulse)	20 10 high or 10 critical 8/ 30 10 low	3.21	4.8.16	1 9/
Vibration, high frequency		3.22	4.8.17	
Group VI				
Life	102 34 high 34 critical 34 low	3.23	4.8.18	1 9/
Group VIA				
Power ratings at 25°C	102 34 high 34 critical 34 low	3.23.1	4.8.18	1 9/
Group VII				
High temperature exposure	20 10 high or 10 critical 8/ 30 10 low	3.24	4.8.19	1 9/
Low temperature storage		3.25	4.8.20	1 9/
Group VIII 4/11/				
Fungus	10	3.26	4.8.21	0

See footnotes on next page.

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1/ Failure of a single network in one or more tests of a group shall be charged as a single defective.
2/ Tests need not be performed if group A inspection has been performed on the qualification samples.
3/ Nondestructive tests.
4/ Sample units for groups II and VIII are not required to be subjected to group I.
5/ Networks shall meet all requirements of group I before subjecting to groups III through VII.
6/ Marking shall be considered defective if the marking is illegible or incorrect.
7/ Tests may be performed on electrical rejects.
8/ When no critical value is specified, only the highest and lowest resistance values shall be tested (20 samples, total).
9/ Number of defective packages allowed per resistance value tested (e.g., low critical and high resistance).
10/ The internal visual examination shall be performed after the final inspection of group IV.
11/ The fungus requirement is either by certification or performance.

4.4.3 Failures. Failures in excess of those allowed in table X shall be cause for refusal to grant qualification.

4.4.4 Failure rate (FR) level and quality level verification.

4.4.4.1 FR qualification. FR qualification shall be in accordance with the general and detailed requirements of MIL-STD-690 and the following details:

a. Procedure I - Qualification at the initial FR level. Level M (1.0 percent) of FRSP-60 shall apply. Sample units shall be subjected to the qualification inspection specified in group V, table X (see 4.4.2). Extended life test sample shall be continued on test to 10,000 hours as specified in 4.8.18, upon completion of the 2,000 hour qualification.

b. Procedure II - Extension of qualification to lower FR levels. To extend qualification to the R (0.01 percent) and S (0.001 percent) FR levels, data from two or more styles of similar construction may be combined.

c. Procedure III - Maintenance of FR levels qualification. Maintenance period A of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification (see 4.7.2.1). For FR levels R and S data from all lead types may be combined.

4.5 Quality level verification. The contractor is responsible for establishing a quality system to verify the ppm defect level of lots that are subjected to subgroup 2 tests of the group A inspections. The ppm defect level shall be maintained for each detail specification. The ppm defect level shall be based on a 6-month moving average. The contractor shall verify and report the individual ppm categories (i.e., ppm-2 and ppm-3) and an overall ppm defect level (i.e., ppm-5) monthly.

4.5.1 Noncompliance. The contractor shall notify the qualifying activity when the 100 ppm level is reached or exceeded for ppm-2. The contractor shall provide sufficient information to the qualifying activity documenting the causes of the problem and what corrective action is being taken. Failure to correct this problem shall be the basis for removal of the affected product from the QPL.

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4.6 Verification of qualification. Every 6 months, the manufacturer shall compile a summary of the results of quality conformance inspections and extended FR test data, in the form of a verification of the qualification report, and forward it to the qualifying activity within 30 days after the end of the reporting period as the basis of continued qualification approval. In addition, the manufacturer shall immediately notify the qualifying activity whenever the FR data indicates that the manufacturer has failed to maintain the qualified FR level, or the group C inspection data indicates failure of the qualified product to meet the requirements of this specification. Continuation shall be based on evidence that over a 6-month period the following has been met:

a. Verification by the qualifying activity that the manufacturer meets the requirements of MIL-STD-790.

b. The manufacturer has not modified the design of the item.

c. The specification requirements for the item have not been amended so far as to affect the character of the item.

d. Lot rejection for group A, subgroup 1 inspection does not exceed 5 percent, or one lot, whichever is greater. (Subgroups 2 and 3, group A, not included in reject rate.)

e. Lot rejection for group B inspection does not exceed 5 percent, or one lot, whichever is greater.

f. The requirements for group C inspection are met.

g. The records of FR tests combined substantiate that the "M" (1.0 percent), or "P" (0.1 percent) FR level has been maintained, or that the manufacturer continues to meet the "R" (0.01 percent), and "S" (0.001 percent) FR level for which qualified, although the total component hours of testing does not, as yet, meet the requirement of 4.4.4.1c.

h. The contractor shall provide documentation to the qualifying activity pertaining to ppm calculations including numbers of part types tested, individual ppm defect categories (i.e., ppm-2 and ppm-3) and the overall ppm defect rate (ppm-5). This information shall be submitted on a detail specification basis.

When group C requirements were not met and the manufacturer has taken corrective action satisfactory to the government, group C retesting shall be instituted. A summary of retesting shall be fowarded to the qualifying activity within 30 days after completion of the retest. All reports are to be certified by a responsible company offical and government inspector.

4.7 Quality conformance inspection.

4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.7.1.1 Inspection and production lot.

4.7.1.1.1 Inspection lot. An inspection lot shall consist of all the networks of similar styles with the same protection enclosure or coating design configuration, and height dimensions manufactured or presented for shipping against this document during a period of not more than a month. Resistors of various resistance values and tolerance may be combined to form a lot provided they are all made with the same enclosure material on the same production line using the same methods and processes. Characteristics shall not be combined.

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4.7.1.1.2 Production lot. A production lot shall consist of all networks of the same style, schematic, nominal resistance value, and resistance temperature characteristic. Manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle.

4.7.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table XI, and shall be made on the same set of sample units, in the order shown.

4.7.1.2.1 Subgroup 1 tests. Subgroup 1 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Networks that are out of resistance tolerance, or which experience a change in resistance greater than that permitted for the tests of this subgroup, shall be removed from the lot. Lots having more than 10 percent total rejects, due to exceeding the specified resistance change limit, shall not be furnished on contracts.

4.7.1.2.1.1 Manufacturer's production inspection. If the manufacturer performs tests similar to those specified in subgroup 1, table XI as the final step of his production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than those specified for subgroup 1. Test conditions shall be equal to or more stringent than those specified for subgroup 1 tests.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criteria are the same or more stringent than those specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests.

f. Once approved, the manufacturer shall not change the test procedures or criteria without prior notification to and concurrence from the qualifying activity.

4.7.1.2.2 Subgroup 2 tests (ppm categories).

4.7.1.2.2.1 Sampling plan. Subgroup 2 tests shall be performed on an inspection lot basis. Samples subjected to subgroup 2 shall be selected in accordance with table XII based on the size of the inspection lot. (Note: Larger samples may be inspected by the contractor in order to calculate ppm. However, rejection of the lot shall be based on one or more defects.) In the event of one or more failures, the lot shall be rejected. Equipment used to perform the subgroup 2 tests shall not be the same as those used in the subgroup 1, 100 percent tests.

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4.7.1.2.2.2 Rejected lots. The rejected lot shall be segregated from new lots and those lots that have passed inspection. The rejected lot shall be 100 percent inspected for those quality characteristics found defective in the sample, and any defects found removed from the lot. A new sample of parts shall then randomly selected in accordance with table XII. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.7.1.2.2.3 PPM calculations. PPM calculations shall be based on the results of the first sample check as prescribed in 4.7.1.2.2.1. Calculations and data exclusion shall be in accordance with EIA-554. (Note: PPM calculations shall not use data on the second sample submission.)

4.7.1.2.3 Subgroup 3 tests. Subgroup 3 shall be performed on an inspection lot basis. Statistical sampling inspection shall be performed on an inspection lot basis. A sample of 13 parts shall then be randomly selected and if one or more defects are found, the lot shall be screened and defects removed. A new sample of 13 parts shall then be randomly selected. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification. Major and minor defects shall be as specified in table XIII.

TABLE XI. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 1			
Thermal shock	3.8	4.8.3	100 percent inspection
Power conditioning	3.9	4.8.4	
DC resistance 1/	3.10	4.8.5	
Subgroup 2 (ppm)			
DC resistance (ppm-2)	3.10	4.8.5	See table XII
Mechanical examination (ppm-3)	3.5	4.8.2	
Subgroup 3			
Visual examination	3.1, 3.4, 3.27	4.8.2	13 samples 0 failures

1/ Networks shall meet the specified initial resistance tolerance. The resistance measurement made upon completion of the power conditioning test may be used if a measurement has been made which can, without conversion, be directly related to nominal resistance value and tolerance.

TABLE XII. Sampling plans for ppm categories.

Lot size	Sample size
0 - 125	100 percent
126 - 3,200	125
3,201 - 10,000	200
10,001 - 35,000	315
35,001 - 150,000	500
150,001 - 500,000	800
500,001 - up	1,250

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TABLE XIII. Classification of defects.

Construction	Requirement	Classification
Body	Cracks or holes which could cause probable failure.	Major
Leads	Leads broken, crushed or nicked which would cause probable failure in use.	Major
	Crushed or nicked at ends which would not cause failure in use.	Minor
	Lead spacing incorrect dimensions.	Major
Marking	Incorrect, illegible marking.	Major

4.7.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table XIV, in the order shown. They shall be performed on sample units that have been subjected to and have passed group A inspection. As an option, subgroup 2 test may be performed on electrical rejects or empty body networks. The empty body networks must be processed through all the standard manufacturing and test flow cycles, oven bakes, and temperature cycles of qualified networks to assure that the network leads and printed surfaces are in the usual condition prior to test.

TABLE XIV. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	Sample size	Number of failures allowed
Subgroup 1				
Resistance-temperature characteristic	3.13	4.8.8	13	0
Dielectric withstanding voltage	3.17	4.8.12		
Insulation resistance	3.18	4.8.13		
Short-time overload	3.15	4.8.10		
Subgroup 2 1/				
Solderability	3.11	4.8.6	6	
Resistance to solvents	3.12	4.8.7	8	0

1/ Tests may be performed on electrical rejects.

4.7.1.3.1 Disposition of sample units. Sample units which have passed all the group B, subgroup 1 inspections may be delivered on the contract provided they are within resistance tolerance and meet the requirements for visual and mechanical inspection.

4.7.2 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.7.2.1.5), delivery of products which have passed groups A and B inspections shall not be delayed pending the results of these periodic inspections.

4.7.2.1 Group C inspection. Group C inspection shall consist of the tests specified in table XV, in the order shown. They shall be performed on sample units that have been selected from lots that have passed groups A and B inspection. Networks of similar styles and method of enclosure may be grouped together. Group C samples shall be representative of production.

4.7.2.1.1 Monthly.

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4.7.2.1.1.1 Subgroup 1. Test samples shall be selected for each inspection lot produced during a 1-month period. These samples shall be accumulated and placed on the life test as specified in 4.8.18, once a month, for the full 10,000-hour life test period. The test sample size shall be determined by the manufacturer so that the unit hours generated meet the requirements specified for the qualified failure rate level (see 4.4.4). In any event, a minimum of five samples shall be selected from each lot. As far as practicable, the resistance values tested during a maintenance period shall be representative of all resistance decades produced during this period. The accumulated data shall be used for maintenance and extention of failure rate qualification.

4.7.2.1.1.2 Subgroup 2. Ten sample units of any resistance value between the critical and low values shall be inspected every month. If none of these values are produced during the month, networks of the lowest resistance value produced shall be inspected, with one defective unit allowed.

4.7.2.1.2 Quarterly. Ten sample units from the highest resistance decade shall be inspected quarterly. Ten sample units from the resistance decade produced shall be subjected to the tests of subgroup 1 with one defective unit allowed.

4.7.2.1.3 Semiannually. Ten sample units from the highest resistance decade produced shall be subjected to the tests of subgroup 1. Ten sample units of any resistance value shall be subjected to the tests of subgroup 2. One defective unit shall be allowed.

TABLE XV. Group C inspection.

Inspection	Number of samples to be inspected	Requirement paragraph	Test method paragraph	Number of failures allowed
Monthly				
Subgroup 1				
Life	4.7.2.1.1	3.23	4.8.18	4.7.2.1.1.1
Subgroup 2				
Low temperature operation	10	3.14	4.8.9	1
Short-time overload		3.15	4.8.10	
Terminal strength		3.16	4.8.11	
Quarterly				
Thermal shock	10	3.8	4.8.3	1
Dielectric withstanding voltage		3.17	4.8.12	
Insulation resistance		3.18	4.8.13	
Resistance to soldering heat 1/		3.19	4.8.14	
Moisture resistance		3.20	4.8.15	
Semiannually				
Subgroup 1				
Shock (specified pulse)	10	3.21	4.8.16	1
Vibration, high frequency		3.22	4.8.17	
Subgroup 2				
High temperature exposure	10	3.24	4.8.19	
Low temperature storage		3.25	4.8.20	

1/ The internal inspection shall be performed after the final inspection of the subgroup.

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4.7.2.1.4 Disposition of samples. Sample units which have been subjected to group C inspection shall not be delivered on the contract.

4.7.2.1.5 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.7.3 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-R-39032.

4.8 Methods of inspection.

4.8.1 Network handling procedures. When specified herein, the networks shall be tested mounted on a test board as described in 4.8.1.1. For those test procedures where mounting requirements are unspecified, the networks may be tested unmounted using pressure type contacts.

4.8.1.1 Specified mounting. When specified in the test procedure, the networks shall be mounted on a suitable test board of glass base nominally .0625 inch (1.59 mm) thick. The test board shall be such that it shall not be the cause of, nor contribute to any failure of network, in any of the tests for which it may be used. Networks must be spaced at least 0.5 inch (1.3 mm) from each other.

4.8.2 Visual and mechanical inspection. Networks shall be inspected to verify that the materials, design, construction, physical dimensions, and workmanship are in accordance with the acceptable requirements (see 3.1, 3.4, 3.5, 3.27, and 3.28).

4.8.3 Thermal shock (see 3.8). Networks shall be tested in accordance with method 107 of MIL-STD-202. The following details and exception shall apply:

a. Mounting: The method used to transport and hold the networks shall ensure circulation of the ambient air to each network.

b. Measurement before cycling: DC resistance shall be measured as specified in 4.8.5.

c. Test condition: For qualification of group I and group A inspection, test condition B. For qualification of group IV and group C inspection, test condition B-1.

d. Measurement after cycling: Within three hours after stabilization at room temperature, dc resistance shall be measured as specified in 4.8.5. For qualification inspection group I, and group A inspection, dc resistance shall be measured following the test of 4.8.4.

Following the test, the networks shall be inspected for evidence of mechanical damage.

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4.8.4 Power conditioning (see 3.9). Networks shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Test temperature and tolerance: 25°C +20°C, -5°C.

b. Initial measurements: Measurements may be made inside or outside the chamber.

(1) Inside chamber: When measurements are to be made inside the chamber, the initial dc resistance shall be measured, at the applicable test temperature, after temperature stabilization, and within 8 hours of exposure of the networks to the test temperature. This initial measurement shall be used as the reference temperature for all subsequent measurements under the same condition.

(2) Outside chamber: When measurements are to be made outside the chamber, the initial dc resistance shall be measured at the room temperature. This initial measurement shall be used as the reference temperature for all subsequent measurements under the same condition.

c. Operating conditions: Rated dc continuous working voltage, filtered or nonfiltered full-wave rectified ac voltage shall be simultaneously applied to each resistor in the network intermittently, 1.5 hours on and 0.5 hour off, for 100 ±4 hours and at the test temperature. During the "on" cycle, the voltage shall be regulated and controlled to maintain 5 percent of the rated continuous working voltage. Unless otherwise specified, power applied shall be 1.5 times rated power (see 3.1).

d. Measurements after test: Following a minimum 0.5 hour stabilization period, dc resistance shall be measured as specified in 4.8.5.

e. Inspection after test: Networks shall be inspected for evidence of mechanical damage.

4.8.5 DC resistance (see 3.10). The dc resistance shall be measured in accordance with method 303 of MIL-STD-202. The following details and exceptions shall apply:

a. Measuring apparatus: The same measuring instrument shall be used for any one test, but not necessarily for all tests.

b. Test voltage: Measurements of resistance shall be made using the test voltages specified in table XVI. The test voltage chosen, whether it be the maximum or a lower voltage which would still provide the sensitivity required, shall be applied across the terminals of the resistor. This same voltage shall be used whenever a subsequent resistance measurement is made.

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TABLE XVI. Standard dc resistance test voltages.

Nominal resistance	Maximum test voltage		
	10 milliwatts	25 to 99 milliwatts	100 to 225 milliwatts
Ohms	Volts	Volts	Volts
10 to 98.8	.3	.5	1.0
100 to 988	1.0	1.0	1.0
1,000 to 9,880	3.0	3.0	3.0
10,000 to 98,800	10.0	10.0	10.0
100,000 or higher	30.0	30.0	30.0

c. Temperature: The dc resistance test specified in group I of table X shall be performed at 25°C ±5°C. For all other tests, unless otherwise specified herein, the temperature at which subsequent and final resistance measurements are made in each test shall be within ±2°C of the temperature at which the initial resistance measurement was made.

d. Procedure: Unless otherwise specified (see 3.1), all resistor elements which can be isolated in the network shall be individually measured. All resistor elements which cannot be isolated in the network shall be measured using a guarded resistance measuring technique as far as practicable (see 3.1). Interpin resistance shall be measured using the test instrument used for measuring the isolated resistors.

4.8.6 Solderability (see 3.11). Networks shall be tested in accordance with method 208 of MIL-STD-202. The following details and exceptions shall apply:

a. All terminals of each network shall be tested.

b. Application of standard solderable wire is not required.

c. Leads shall be solderable .010 inch (0.25 mm) above seating plane.

d. Flat packs, the seating plane will be defined as 0.060 inch (1.52 mm) from component body, excluding any epoxy meniscus.

4.8.7 Resistance to solvents (see 3.12). Networks shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

a. The marked portion of the network body shall be brushed.

b. The number of sample units shall be as specified in tables X and XIV, as applicable.

c. Networks shall be inspected for mechanical damage and legibility of markings.

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4.8.8 Resistance-temperature characteristic (see 3.13). Networks shall be tested in accordance with method 304 of MIL-STD-202. The following details and exceptions shall apply:

a. Reference temperature: Room ambient temperature.

b. Test temperature: In accordance with table XVII.

TABLE XVII. Resistance-temperature characteristic.

Sequence	Temperature	
	Qualification inspection	Group B acceptance inspection
	°C	°C
1	Room temperature 1/	Room temperature 1/
2	-15 ±3	-55 ±3
3	-55 ±3	Room temperature 1/
4	Room temperature 1/	125 ±3
5	65 ±3	---
6	125 ±3	---

1/ This temperature shall be considered the reference temperature for each of the succeeding temperatures.

4.8.9 Low temperature operation (see 3.14). Following a dc resistance measurement as specified in 4.8.5, the networks shall be placed in a cold chamber at -65°C +0°C, -5°C. After 1 hour of stabilization at this temperature, full rated continuous working voltage as specified in 3.7 shall be applied for 45 minutes. The networks may be loaded individually or in parallel. Fifteen +5, -0 minutes after the removal of the voltage, the temperature in the chamber shall be gradually increased to room temperature within a period of not more than 8 hours. The networks shall be removed from the chamber and maintained at a temperature of 25°C ±5°C for a period of approximately 24 hours; the dc resistance shall then be measured as specified in 4.8.5. Networks shall then be inspected for evidence of mechanical damage.

4.8.10 Short-time overload (see 3.15). The dc resistance shall be measured as specified in 4.8.5. A dc test potential, 2.5 times the rated continuous working voltage but not exceeding twice the maximum voltage (see 3.1), shall be applied for 5 ±1 seconds to each resistor in the network, one at a time. If single resistors cannot be isolated, the voltage applied at the two terminals shall be $2.5\sqrt{PR}$, where R is the actual resistance measured at the two terminals, and P is the combined power rating of the elements contributing to the measured resistance. The following conditions shall be maintained.

a. Networks are to be mounted horizontally in still air with no circulation other than that created by the heat of the network being operated.

b. Ambient temperature during test shall be 25°C ±5°C.

c. Thirty +15, -0 minutes after removal of the test potential, the resistance shall be measured as specified in 4.8.5. Networks shall be inspected for evidence of arcing, burning, and charring.

d. For reference purposes, a 10 second waiting period between voltage pulses shall be used.

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4.8.11 Terminal strength (see 3.16).

4.8.11.1 Pull test. Networks shall be tested as follows:

a. Test weight:

(1) 24 ounces for flat packs.

(2) 4-1/2 pounds for other body styles.

b. Test time: 30 seconds.

c. Number of terminals to be tested: 5 randomly selected terminals.

d. Procedure: The specified weight shall be applied, without shock, to each lead or terminal to be tested in a direction parallel to the axis of the lead (or terminal) and maintained for the specified test time. The tension shall be applied as close to the end of the lead (or terminal) as practicable.

e. Measurement after test: Networks shall be inspected for evidence of mechanical damage, and resistance shall be measured as specified in 4.8.5.

4.8.11.2 Bend test.

4.8.11.2.1 Bend test for configurations other than flat packs. Networks in dual in-line and single in-line packages shall be tested as follows:

a. Number of terminals to be tested: 5 randomly selected terminals.

b. Procedure: Each lead, selected to be tested, of the network shall be bent through an angle sufficient to cause the lead to retain a permanent bend (i.e., after stress removal) of at least 15 degrees measured at the lead extremities about the first bend. At the completion of the initial bend, the leads shall be returned to their approximate original position. This procedure shall be performed three times.

c. Measurement after test: Networks shall be inspected for evidence of mechanical damage, and resistance shall be measured as specified in 4.8.5.

4.8.11.2.2 Bend test for flat packs. Networks in flat pack shall be tested as follows:

a. Number of terminals to be tested: 5 randomly selected terminals.

b. Procedure: Unless otherwise specified (see 3.1), a force of 8 ±0.5 ounces shall be applied to each lead to be tested for three 90 ±5 degree arcs of the case. For leads with a section modulus equal or less than that of a lead with a cross-section of 0.006 x 0.020 inch, the force shall be 3 ±0.3 ounces. An arc is defined as the movement of the case, without torsion, to a position perpendicular to the pull axis and returned to normal. All arcs on a single lead shall be made in the same direction and in the same plane without lead restriction. A bending cycle shall be completed in 2 to 5 seconds. For networks with rectangular or ribbon leads, the plane of the arcs shall be perpendicular to the flat plane of the lead. The test shall not be applied to end leads of packages where its application will apply primarily torsion forces at the lead seal.

c. Measurement after test: Networks shall be inspected for evidence of mechanical damage, and resistance shall be measured as specified in 4.8.5.

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4.8.12 Dielectric withstanding voltage (see 3.17).

4.8.12.1 Atmospheric pressure. Networks shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Mounting:

(1) Dual in-line packages (DIP): The network shall be clamped by mounting with its top on a metal plate of sufficient size to extend beyond the network extremities. An additional metal plate shall be mounted against the bottom of the network, avoiding contact with the terminals. The terminals of the network shall be connected together.

(2) Single in-line (SIP) and flat packages: The network shall be clamped by mounting with its sides on metal plates of sufficient size to extend beyond the network extremities and the terminals shall be connected together.

b. Magnitude of test voltage: 200 volts rms for DIPs and SIPs, and 100 volts for flat packs.

c. Nature of potential: An ac supply at commercial-line frequency (not more than 100 hertz (Hz)) and waveform.

d. Duration of application of test voltage: 1 minute.

e. Points of application of test voltage: Between the terminals connected together and metal-mounting plate.

f. Inspections and measurements: During the tests, the leakage current shall be monitored and the networks inspected for evidence of arcing and breakdown. At the conclusion of the test, networks shall be inspected for evidence of damage.

4.8.13 Insulation resistance (see 3.18). Networks shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

a. Test condition A.

b. Special preparation: As specified in 4.8.12.1a.

c. Points of measurement: As specified in 4.8.12.1e.

4.8.14 Resistance to soldering heat (see 3.19). Networks shall be tested in accordance with method 210 of MIL-STD-202. The following details shall apply:

a. Measurement before test: DC resistance shall be measured as specified in 4.8.5.

b. Special preparation of specimen: Sample units shall not have been soldered during any of the previous tests.

c. Test condition C (solder 260°C, immersion 10 seconds).

d. Depth of immersion shall be .0625 inch (1.59 mm) from seating plane.

e. Measurement after test: After completion of terminal immersion and following a minimum 3-hour cooling period, dc resistance shall be measured as specified in 4.8.5.

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f. Inspection after test: Networks shall be inspected for evidence of mechanical damage.

g. Internal inspection: Upon completion of the final inspection of group IV in qualification inspection and quarterly, subgroup 1 of group C inspection, the networks shall be decapped and inspected for evidence of solder reflow and heat damage.

4.8.15 Moisture resistance (see 3.20). Networks shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Initial measurement: Immediately following the initial conditioning period, dc resistance of resistors in the network shall be measured as specified in 4.8.5.

b. Loading voltage: A dc voltage equivalent to 0.10 of rated wattage shall be applied to all resistor elements during the first 2 hours of steps 1 and 4.

c. Subcycle: Step 7b shall not be applicable. Step 7a shall be performed during any five of the first eight cycles only.

d. Measurements at high humidity: None.

e. Final measurements: Upon completion of step 6 of the final cycle, the resistor networks shall be removed from the chamber and within 24 hours, without any additional handling, the dc resistance, dielectric withstanding voltage, and insulation resistance shall be measured in accordance with 4.8.5, 4.8.12, and 4.8.13. When the networks are tested as specified, the change in resistance between initial and final measurements shall not exceed the value specified in 3.20.

f. Inspection after test: Networks shall be inspected for evidence of mechanical damage.

4.8.16 Shock (specified pulse) (see 3.21). Networks shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

a. Special mounting means: Networks shall be mounted in accordance with 4.8.1.1 on an appropriate mounting fixture. The mounting fixture shall be constructed in such a manner as to insure that the mounting supports remain in a static condition with reference to the shock-test table. Networks shall be mounted in relation to the test equipment in such a manner that the stress applied is in the direction which would be considered most detrimental. Single-in-line networks with a height profile in excess of .200 inch (5.08 mm) (styles RZR40, RZR50, and RZR60), shall be mounted as specified. However, the body of the network shall be cemented to the printed circuit board. The cement material shall not extend up the body more than 50 percent in the vertical plane or beyond the resistor body ends in the horizontal plane. In no case shall the resistor body be completely encapsulated (see figure 3).

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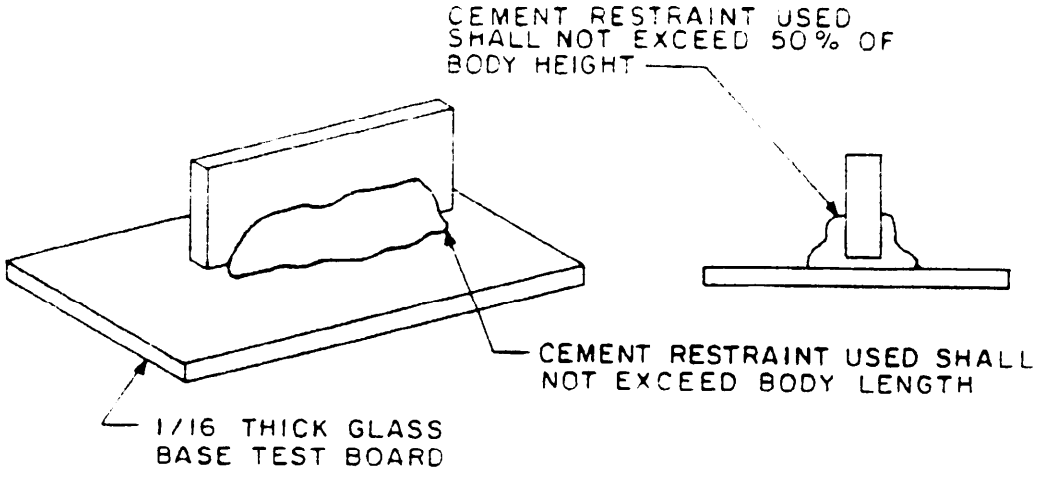


FIGURE 3. Special mounting for high profile SIPS.

b. Test leads: Test leads used during this test shall be no larger than AWG size 22 stranded wire, so that the influence of the test lead on the resistor shall be held to a minimum. The test-lead length shall be no longer than necessary.

c. Measurements before shock: Resistance shall be measured as specified in 4.8.5.

d. Test condition I.

e. Measurements after shock: Resistance shall be measured as specified in 4.8.5.

f. Inspection after shock: Networks shall be inspected for evidence of mechanical damage.

4.8.17 Vibration, high frequency (see 3.22). Networks shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: As specified in 4.8.16a.

b. Test leads: As specified in 4.8.16b.

c. Measurements before vibration: As specified in 4.8.16c.

d. Test condition D.

e. Measurements after vibration: As specified in 4.7.16e.

f. Inspection after vibration: Networks shall be inspected for evidence of mechanical damage.

4.8.18 Life (see 3.23). Networks shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: As specified in 4.8.1.1, with wiring to be connected to landing pads on the top and bottom of the circuit board. If forced air circulation is employed, the air velocity shall not exceed 500 feet per minute and there shall be no direct impingement of the forced-air supply upon the networks.

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b. Test temperature and tolerance:

(1) 70°C power rating: 70°C ±5°C.
Qualification inspection: 25°C ±5°C.

(2) Failure rate level determination: 125°C, +15°C, -5°C.

c. Initial measurements: Measurements may be made inside or outside the chamber.

(1) Inside chamber: When measurements are to be made inside the chamber, the initial dc resistance shall be measured at the applicable test temperature, after temperature stabilization, and within 8 hours of exposure of the resistor networks to the test temperature. This initial measurement shall be used as the reference temperature for all subsequent measurements under the same condition.

(2) Outside chamber: When measurements are to be made outside the chamber, the initial dc resistance shall be measured at the room temperature. This initial measurement shall be used as the reference temperature for all subsequent measurements under the same condition.

d. Operating conditions: Rated dc continuous working voltage, filtered or nonfiltered full-wave rectified ac voltage shall be simultaneously applied to each resistor in the network intermittently, 1.5 hours on and 0.5-hour off, for the specified number of hours and at the test temperature. During the "on" cycle, the voltage shall be regulated and controlled to maintain the rated continuous working voltage of 5 percent. In no case shall the network power rating be exceeded.

e. Test condition: 2,000 hours elapsed time for qualification inspection with all samples continued to 10,000 hours; 2,000 hours for 25°C power rating test; and 10,000 hours for failure rate level inspection group C.

f. Measurements during test:

(1) DC resistance shall be measured at the end of the 0.5-hour off periods after 250 +48, -0; 500 +48, -0; 1,000 +48, -0, and 2,000 +72, -0 hours have elapsed hours.

(2) Extended life testing: DC resistance shall be measured at the end of the 0.5-hour off periods after 250 +48, -0; 500 +48, -0; 1,000 +48, -0; and 2,000 +72, -0 hours and every 2,000 +96, -0 thereafter until the required 10,000 hours has elapsed. Measurements shall be made as near as possible to the specified time but may be adjusted (within specified tolerances) so that measurements need be made only during the normal workday.

(3) Measurements outside of chamber: When measurements are made outside the chamber, networks shall be outside of the chamber for a minimum of 45 minutes and stabilized before measurement.

g. Inspection after test: Networks shall be inspected for evidence of mechanical damage.

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4.8.19 High temperature exposure (see 3.24).

a. Mounting: Not required.

b. Initial measurements: DC resistance shall be measured as specified in 4.8.5 at 25°C ±2°C.

c. Procedure: Following initial resistance measurements, networks shall be placed in a chamber with forced air circulation maintained at 125°C ±5°C for a period of 100 ±4 hours with no load applied.

d. Final measurements: After removal from the test chamber, networks shall be permitted to stabilize at an ambient temperature of 25°C ±5°C for 4 hours. Resistance measurements shall be made as specified in 4.8.5. Networks shall be inspected for evidence of mechanical damage.

4.8.20 Low temperature storage (see 3.25).

a. Mounting: Not required.

b. Procedure: DC resistance shall be measured as specified in 4.8.5. Within 1 hour after this measurement, the networks shall be placed in a cold chamber at -65°C +0°C, -3°C. Twenty-four hours after the networks have reached this temperature, the temperature of the chamber shall be gradually increased to room temperature within a period of not more than 8 hours. The networks shall be removed from the chamber and maintained at a temperature of 25°C ±5°C for a period of approximately 24 hours; the dc resistance shall then again be measured as specified in 4.8.5. Networks shall then be inspected for evidence of mechanical damage.

4.8.21 Fungus (see 3.26). Networks shall be tested in accordance with method 508 of MIL-STD-810. Networks shall be inspected for evidence of fungus.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Networks described herein are intended to be used in electronic circuits where miniaturization is required.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL874 whether or not such products have actually been so listed by that date. The attention of the contractor is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45444-5000.

6.4 Application notes.

6.4.1 Stacking of networks. Stacking is generally not recommended, but if required, care should be taken to compensate for the lower heat dissipation capabilities by derating the wattage rating.

6.4.2 Close-tolerance networks. Close tolerance networks (i.e., 0.1 percent and tighter) should be mounted by a method which produces the least heating effect over a short time to avoid permanent change in resistance.

6.4.3 Caution note. When the printed wiring assembly utilizes components made of brittle materials (glass or ceramic), such components should be protected, prior to coating, against breakage by the conformal coating type covering the component with transparent, clean, thin, pliant buffer material securely fitted, such as with heat shrinkable sleeving, polyethylene terephthalate covered by MIL-I-23053/7. The buffer material should cover the entire component and should not extend over the component by more than 0.062 inch (1.57 mm) and should be compatible with the conformal coating material. Buffer material may not be needed when types SR and XY in accordance with MIL-I-46058 are used.

6.5 Power ratings at 25°C. Power ratings for networks under this specification have been established by a 2,000 hour life test, and a ΔR = ±0.50 percent for characteristics H and K, and a ΔR = ±2.0 percent for characteristic M. The power ratings established should be in accordance with figure 1.

6.6 Subject term (key word) listing.

DIP
Flat pack
Network
Resistor
SIP

6.7 Part or Identifying Number (PIN). PIN is a new term encompassing previous terms used in specifications such as part number, type designator, identification number etc. (see 1.2.1).

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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 Scope. This appendix details the procedure for submission of samples, with related data, for qualification inspection of networks covered by this specification. The procedure for extending qualification of the required sample to other networks covered by this specification is also outlined herein. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. SUBMISSION

30.1 Sample. A sample consisting of 108 sample units of each resistance value (lowest, highest, and critical or nearest to critical value (see table XVIII)) and in each style, characteristic, resistance tolerance, and schematic for which qualification is sought, shall be submitted and subjected to the inspection of table X. Twelve sample units of any value shall be submitted and subjected to the tests of group II of table X. In addition, 10 sample units of any characteristic or resistance value shall be submitted and subjected to the test of group VIII of table X. The range extension at higher or lower values will be qualified by supplemental testing of the new highest or lowest value in accordance with the following: Submit 60 additional samples of the new resistance value to be qualified to groups I and IA of table X; of this group, 10 samples shall be subjected to group III, 10 samples to group IV, 10 samples to group V, 10 samples to group VI and 10 samples to group VII. An additional sample submission is required under the following conditions:

a. Qualification to a closer tolerance than submitted above is desired:

Submit:

30 additional samples of the closer tolerance. To be qualified to group I of table X.	10 high value 10 critical value 10 low value

b. Qualification to "J" or "H" schematics by schematics A, B, C, or G submission is desired:

Submit:

40 additional samples of schematic "J" or "H". All to group I, and 10 each to groups III, IV, VI, and VII.	40 highest value

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30.2 Test data. When inspections and tests are to be performed at a Government laboratory, prior to submission, all sample units shall be subjected to all of the tests indicated as nondestructive in table X. Each submission shall be accompanied by the test data obtained from these tests. The performance of the destructive inspections and tests by the manufacturer on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.

30.3 Description of items. The manufacturer shall submit a detailed description of the network being submitted for inspection, including materials used for the resistance element and the protective enclosure or coating. For characteristics C and V, the manufacturer shall submit a description of the precap visual examination used for inspection of: (1) internal connections, (2) metallization, (3) die mounting, and (4) inspection for foreign or extraneous material. After qualification has been granted, no changes shall be made in materials, design, or construction without proper notification to the qualifying activity.

40. EXTENT OF QUALIFICATION

40.1 Extension of qualification. The resistance range included in the qualification of any one network style will be between the lowest and highest resistance values which pass the qualification inspection (see 3.1). Qualification of one characteristic is basis for qualification of another characteristic, as indicated in table XIX. Qualification of the lower resistance tolerances will qualify for the higher resistance tolerances in accordance with table XX. Extent of qualification by style shall be in accordance with table XXII. Extent of qualification by schematic shall be in accordance with table XXII. As a requisite for extension of qualification as described herein between characteristics, tolerances, schematics, resistance values and failure rates, the product involved must be manufactured using the same facilities, processes, and materials as the product originally submitted for qualification.

40.2 Extension of qualification based on MIL-R-83401. The manufacturer may be qualified to this specification based, in part, upon qualification to the nonestablished reliability specification, MIL-R-83401. The manufacturer only has to perform the group I and group VI (life tests) provided the following criteria is met:

a. The actual part design is the same as that qualified to MIL-R-83401 for the specific style, package, resistance range, resistance-temperature characteristic, resistance tolerance and schematics.

b. The parts are made on the same production lines, using the same materials and processes as those qualified to MIL-R-83401.

c. The manufacturer's product has been subjected and passed all requirements of the MIL-STD-790 program.

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APPENDIX

TABLE XVIII. Critical resistance value for qualification inspection. 1/

Style	Critical resistance value 1/ 2/					
	Schematic A	Schematic B	Schematic C	Schematic G	Schematic J	Schematic H
				Characteristics		
	H,K,M MΩ	H,K,M MΩ	H,K,M MΩ	H,K,M MΩ	H,K,M MΩ	H,K,M MΩ
RZR10	0.047	0.1			.200	
RZR20	0.047	0.1			.200	
RZR30	0.047	0.1			.160	
RZR40			.012	.012		.020
RZR50			.012	.012		.020
RZR60			.012	.012		.020
RZR70			.020	.020		.033
RZR80			.020	.020		.033
RZR90			.020	.020		.033
RZR95	0.047	0.10			.160	

1/ Maximum continuous working voltage shall be applied (see 3.1).
2/ The critical resistance value is the maximum standard resistance value which will dissipate full wattage when maximum continuous working voltage is applied.

TABLE XIX. Extent of qualification of characteristics.

Characteristic submitted	Characteristic qualified
H	H,K,M
K	K,M
M	M

TABLE XX. Extent of qualification of resistance tolerance.

Resistance tolerance submitted	Resistance tolerance qualified
B	B, D, F, G, J
D	D, F, G, J
F	F, G, J
G	G, J
J	J

TABLE XXI. Extent of qualification by style.

Style	Will qualify style(s)
RZR10	RZR10
RZR20	RZR10, RZR20,
RZR30	RZR30,
RZR40	RZR40
RZR50	RZR40, RZR50
RZR60	RZR40, RZR50, RZR60
RZR70	RZR70
RZR80	RZR70, RZR80
RZR90	RZR70, RZR80, RZR90
RZR95	RZR30

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APPENDIX

TABLE XXII. Extent of qualification by schematic.

Schematic	Will qualify schematic(s) 1/
A	A, J
B	B, A, J, K, M, N
C	C, G, H
G	G, H

1/ When H or J schematics are included, additional test samples must be tested as specified in 30.1b.

40.3 Extension of qualification by failure rate levels. The extension of qualification by failure rate levels shall be as specified in table XXIII.

TABLE XXIII. Extension of qualification by failure rate levels.

Failure rate level	Will qualify failure rate level
S	S, R, P, M
R	R, P, M
P	P, M
M	M

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 11, 17
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1154)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-874	Resistor Networks, Fixed, Film, Established Reliability, General Specification For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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